Applied Machine Learning



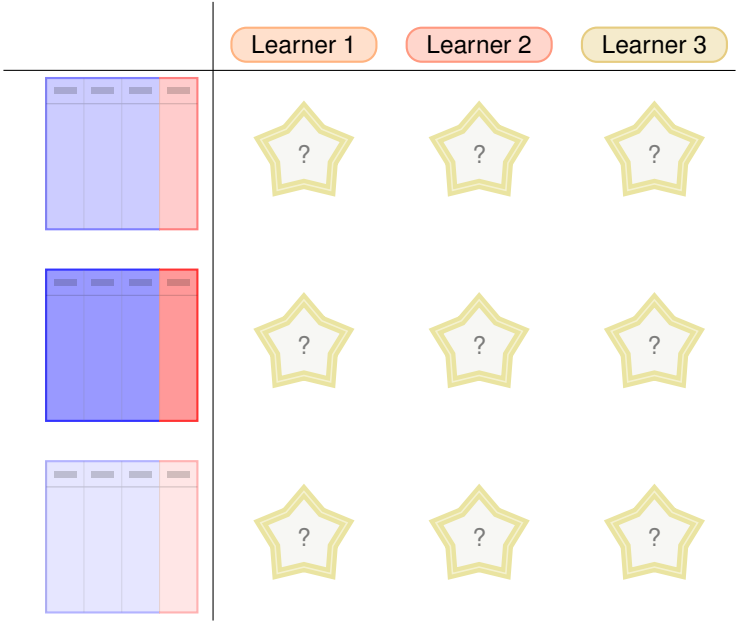
Machine Learning in R
MLR3 Benchmarking & Summary



Learning goals

- Setting up benchmarking experiments
- Working with ‘BenchmarkResult‘ object
- Parallelization
- MLR3 resources and help

PERFORMANCE COMPARISON





PERFORMANCE COMPARISON



- Multiple Learners, multiple Tasks:

```
library("mlr3learners")
learners = list(lrn("classif.rpart"), lrn("classif.kknn"))
tasks = list(tsk("iris"), tsk("sonar"), tsk("wine"))
```

PERFORMANCE COMPARISON



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- Set up the *design* and execute benchmark:

```
design = benchmark_grid(tasks, learners, cv5)
bmr = benchmark(design)
```

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```

- We get a `BenchmarkResult` object which shows that `kknn` outperforms `rpart`:

```
bmr_ag = bmr$aggregate()
bmr_ag[, c("task_id", "learner_id", "classif.ce")]
```

```
#>    task_id    learner_id classif.ce
#>     <char>        <char>      <num>
#> 1:    iris classif.rpart      0.053
#> 2:    iris  classif.kknn      0.040
#> 3:   sonar classif.rpart      0.274
#> 4:   sonar  classif.kknn      0.130
#> 5:    wine classif.rpart      0.157
#> 6:    wine  classif.kknn      0.039
```

BENCHMARK RESULT



What exactly is a `BenchmarkResult` object?
Just like `Prediction` and `ResamplingResult`!

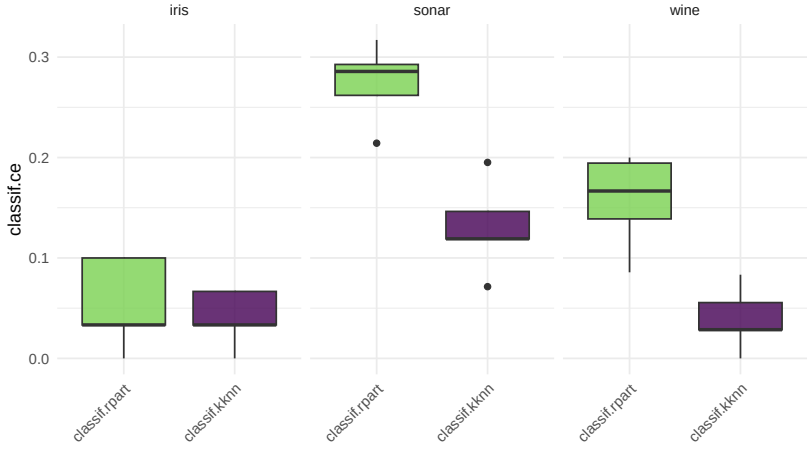
- Table representation using `as.data.table()`
- Active bindings and functions that make information easily accessible

BENCHMARK RESULT

The `mlr3viz` package contains `autoplot()` functions for many mlr3 objects



```
library(mlr3viz)
autoplot(bmr)
```



CONTROL OF EXECUTION



Parallelization

```
future::plan("multicore")
```

- runs each resampling iteration as a job
- also allows nested resampling (although not needed here)

Encapsulation

```
learner$encapsulate = c(train = "callr", predict = "callr")
```

- Spawns a separate R process to train the learner
- Learner may segfault without tearing down the session
- Logs are captured
- Possibilty to have a fallback to create predictions



Help and Summary

HOW TO GET HELP



- Where to start?
  - Check these slides
  - **Check the mlr3book https://mlr3book.mlr-org.com**

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- Where to start?
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  - **Check the mlr3book https://mlr3book.mlr-org.com**
- Get help for R6 objects?
  1. Find out what kind of R6 object you have:

```
class(bmr)
#> [1] "BenchmarkResult" "R6"
```

  2. Go to the corresponding help page:

```
?BenchmarkResult
```

New: open the corresponding man page with

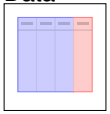
```
learner$help()
```

OVERVIEW



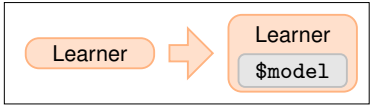
Ingredients:

Data



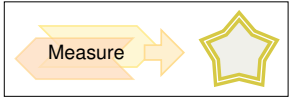
`TaskClassif`,
`TaskRegr`,
`tsk()`

Learning Algorithms



`lrn()` ⇒ `Learner`,
↪`Learner$train()`,
↪`Learner$predict()` ⇒ `Prediction`

Performance Evaluation



`rsmp()` ⇒ `Resampling`,
`msr()` ⇒ `Measure`,
`resample()` ⇒ `ResamplingResult`,
↪ `ResamplingResult$score()`,
↪ `ResamplingResult$aggregate()`

Performance Comparison



`benchmark_grid()`,
`benchmark()` ⇒ `BenchmarkResult`